AI COMPUTING CONTINUUM



AI Computing Continuum

Extending the Open Data Center Ecosystem Across Cloud, Edge, and Industry



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Vincent Nguyen *Fellow HPE*

Dirk Blevins *Senior Director Intel Datacenter Group*

Presenters



Masahisa Kawashima

NTT, Inc. & IOWN Global Forum Chair



Vincent Nguyen

Fellow, HPE



Dirk Blevins

Sr. Director, Intel Data Center Group

Hyperscale Open Data Center Eco-system

OPEN BUILDING BLOCK EXAMPLES

OCP MHS
Modular Hardware System — shared host processor modules & common form factors

OCP DC-SCM
Secure Control Module — decoupled management & hardware root of trust

OCP AI System Architecture
Open rack, power, cooling & interconnect specs for AI clusters

Open Rack ORV3 + Power
48V power shelf, liquid cooling and blind-mate rack infrastructure

AI RACK PLATFORM EXAMPLES



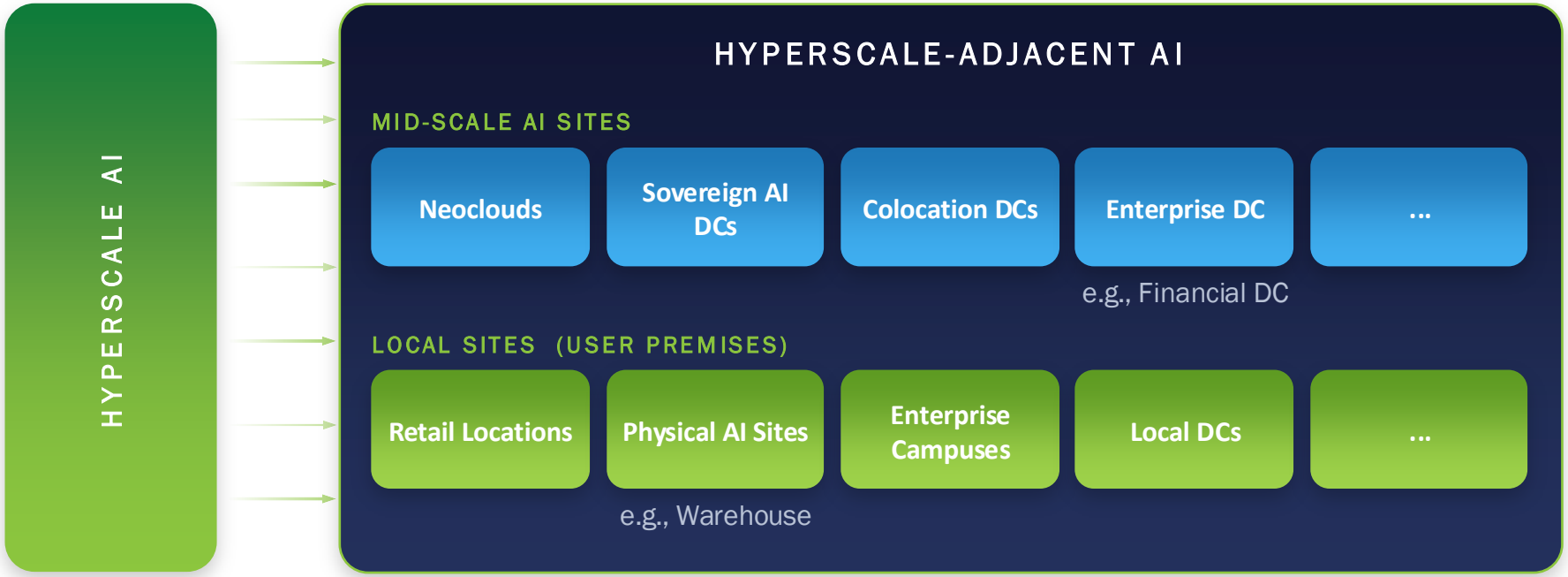
SCALE-UP
INSIDE THE RACK
Coherent accelerator fabric — NVLink / UALink domains within one rack

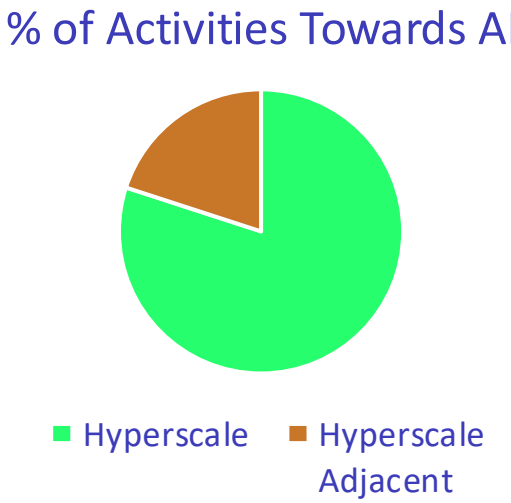
SCALE-OUT
ACROSS THE CLUSTER
Ethernet / UEC & RoCE fabrics linking thousands of racks into a pod

SCALE-ACROSS
BEYOND THE DATA CENTER
Multi-site AI — DC-to-DC and edge continuum interconnect

One Common Open Datacenter Ecosystem

Hyperscale-Adjacent Scope



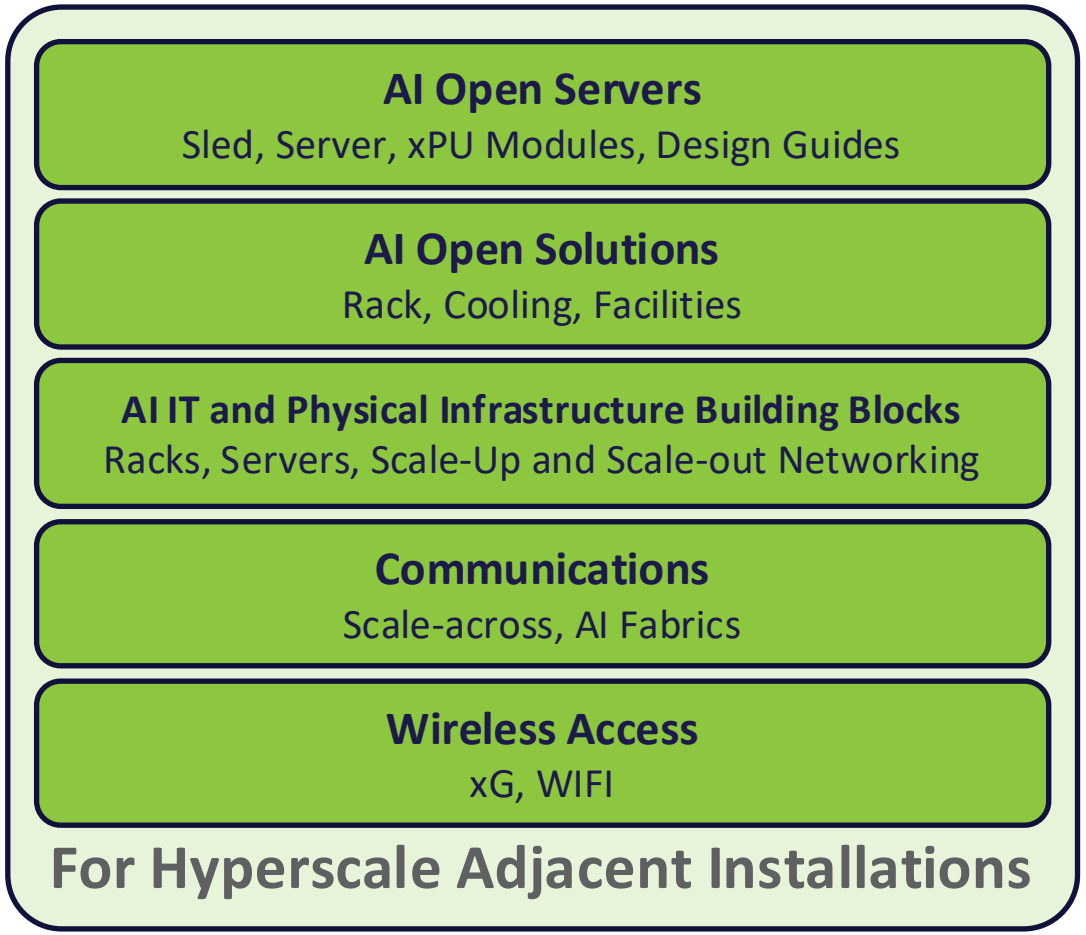


From AI PC to AI Infrastructure: Platform Contracts for Scalable CPU–Accelerator Systems

Tuesday, October 13, 2026 | 4:15 PM PDT | SJCC, Concourse Level, 211

Dong Wei (Arm) | Suraj Varma (Arm, Inc.)

AICC Core Principles



- Scale as you go. Single System to Racks to Rows
- Flexible or Constrained Power Budgets
- Interoperable, Open and Manageable
- Designed with Sustainability in mind
- Scale -Up -Out -Across

Two Years of x86 Ecosystem Collaboration: Open Infrastructure, AI Readiness, and Developer Predictability

Tuesday, October 13, 2026 | 2:20 PM PDT | SJCC, Concourse Level, 211

Jeff McVeigh (Intel) | Stephen Watt (Red Hat) | Andrew Wheeler (HPE) | Martin Dixon (Google) | Jon Lange (Microsoft) | Robert Hormuth (AMD)

AICC Challenges to Address

Location	**Cloud Public**	**Distributed Datacenter (DC) – Private or Public**	**Access Semi-Private**	**Edge / On-Prem Private**
Customer Type	Hyperscale / Large Enterprise	Small/Med/Large Enterprise, NeoCloud, CoLo	CoSPs, CoLo, PoP	Retail Outlets, Quick Serve, Industrial, Transportation, etc.

Rack Scale Architecture

Different servers with different management and security





Nothing in between

4x GPU with no Scale Up
1.5 - 5KW



Sovereign AI
Need for isolation
(no multi-tenancy required)

Some Deployments

Hyperscale Adjacent

8x GPU Scale Up
8 - 15KW



Scale Up
(OAC targeted)

1x GPU no Scale Up
0.8KW - 2KW
Short Depth



1x to 4x GPU, no Scale Up
3KW - 5KW
Full Length



Ruggedized
Liquid Cooled



Limited thermal & power delivery capacity

1x GPU no Scale Up
0.8KW - 3KW
Short Depth

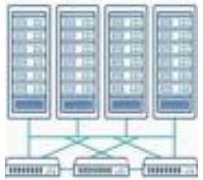


Security Critical

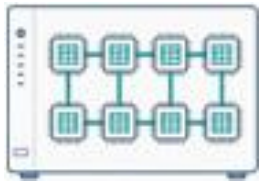
Scale Across



Scale Out



Scale Up



Hyperscale Adjacent Means For

AI-Native Edge MHS 1.0 | Tue, Oct 13 | 3:15 PM PDT
SJCC Concourse Level 211 - OCP Global Summit 2026, San Jose

Scale in AICC Distributed Deployments

The Open Adapter Card - Light

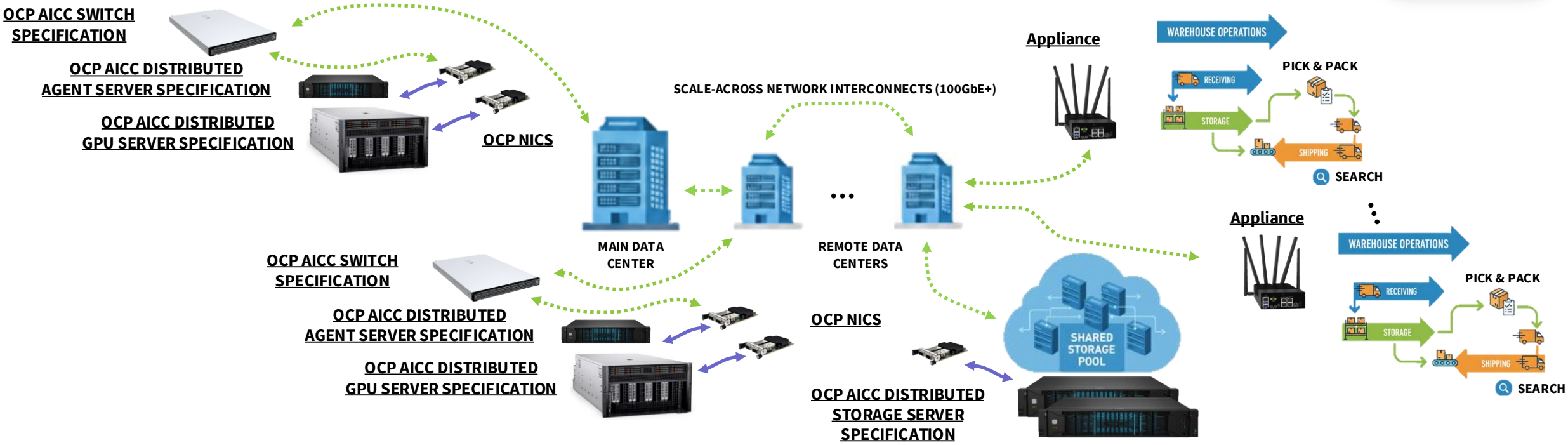
Tuesday, October 13, 2026 | 3:35 PM PDT | SJCC, Concourse Level, 211

Dirk Blevins (Intel) | Vincent Nguyen (Hewlett Packard Enterprise)

Brownfield Deployments

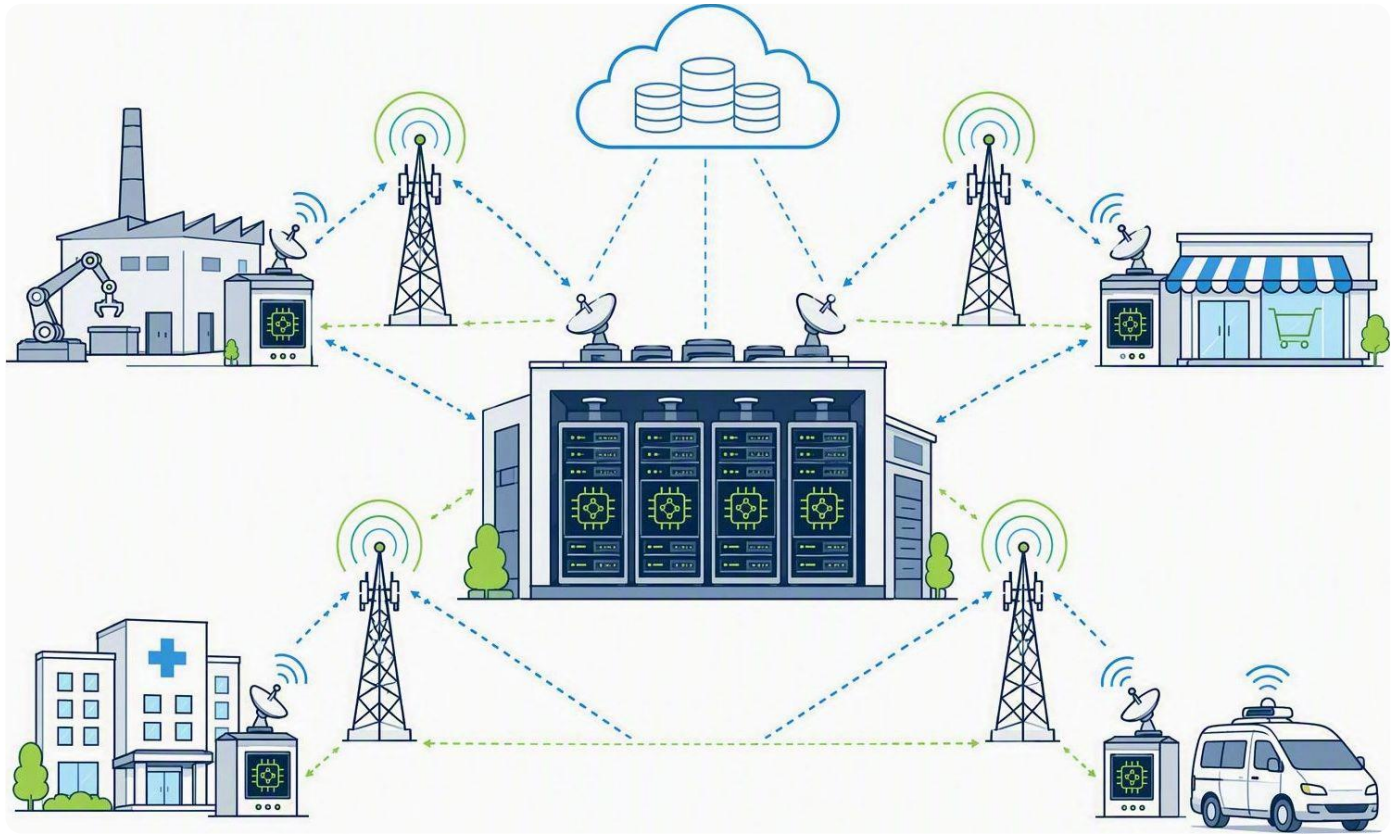
- Too Heavy
- Too Much Power
- Too Hard to cool
- Not Adjacent
- Distributed Locations
- Rack & Stack (Pay-as-you-grow)





Everything Can’t Be Wired

OCP AICC AI Native Wireless: 5G and 6G broadband links distributed AI compute where fiber, power, and floor space cannot reach



Wireless scale-across

5G and 6G backhaul stitches distributed AI nodes together when a single wired hall is not an option.

AI-native air interface

6G is designed with AI in the RAN - inference runs at the tower, not only across it.

Reach the non-adjacent

Factories, retail, clinics and vehicles join the AI fabric over licensed and private spectrum.

Pay-as-you-grow

Add radio-connected nodes incrementally - no heavy rack-and-stack rebuild at each site.

Tuesday, October 13, 2026 | 3:55 PM PDT | AI Computing Continuum track | OCP Global Summit 2026, San Jose (Oct 12-15)

The Autonomous AI-RU: Closing the Loop from RIC Intelligence to the Radio through an Open Radio Abstraction Interface

AI Computing Continuum

Project Leads
Dirk Blevins, Vincent Nguyen

Delivering on the Next Wave of AI: Inference

AI Computing Continuum aims to accelerate the adoption **of scalable, interoperable, and sustainable AI systems beyond traditional hyperscale data centers** by defining open, modular infrastructure standards that work across diverse **hyperscale adjacent** environments—including regional colocation, enterprise on premises, telco points of presence, and industrial sites. Guided by OCP principles of openness, efficiency, and sustainability, the project focuses on leveraging hyperscale data center innovations to develop community-driven, hardware-focused specifications that **enable hyperscale adjacent scale AI deployments covering scale up, scale out, scale across, and wireless access communications.** By standardizing core platform building blocks, bridging the gap between hyperscale and enterprise needs, and addressing real-world constraints in power, thermal, and physical footprints, the AI Computing Continuum initiative supports repeatable, cost-effective deployment of advanced AI compute systems wherever they are needed outside of the centralized hyperscale data centers.

Active Sub-Projects

AI-Native Wireless Access

Project Leads
Brad Deforge, Bradford Stimpson, Thomas Nadeau

AI-Native Edge-MHS

Project Leads
David Heinrich, Brian Baker

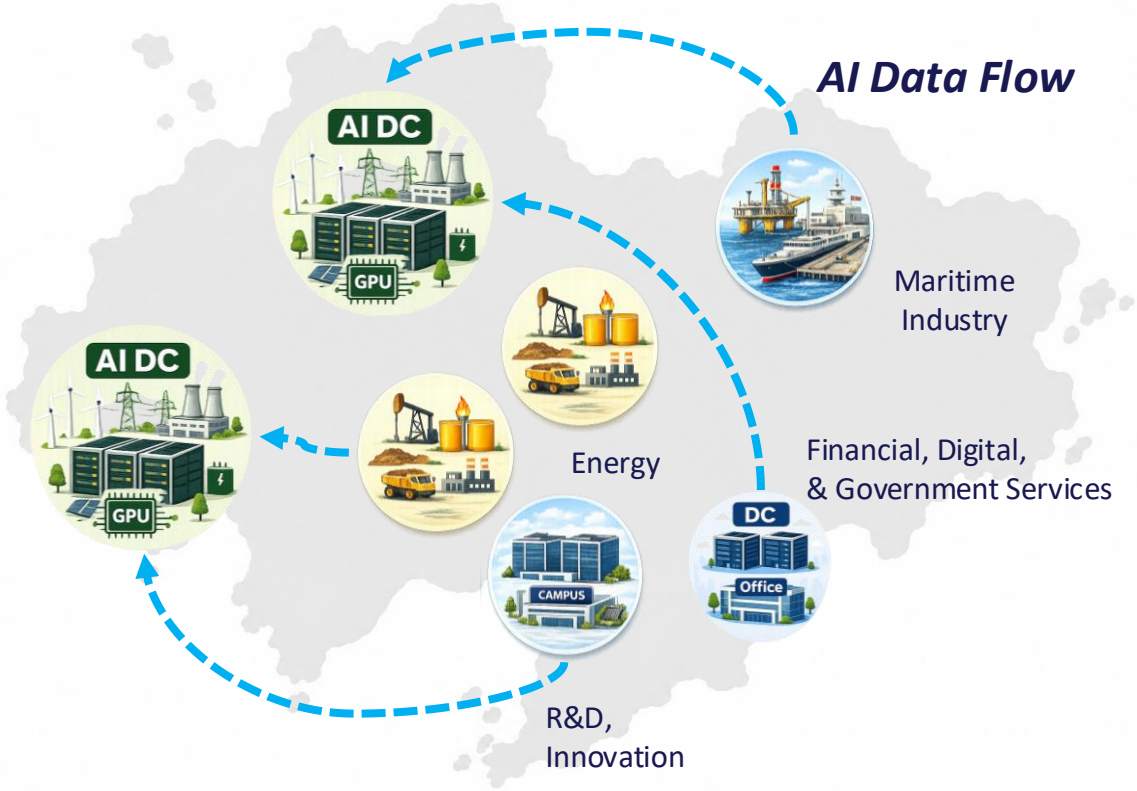
AICC from IOWN Global Forum's Perspective

Why AICC

- From developing AI to **using AI**
- Demand for **sovereignty**
- Data Center moratorium

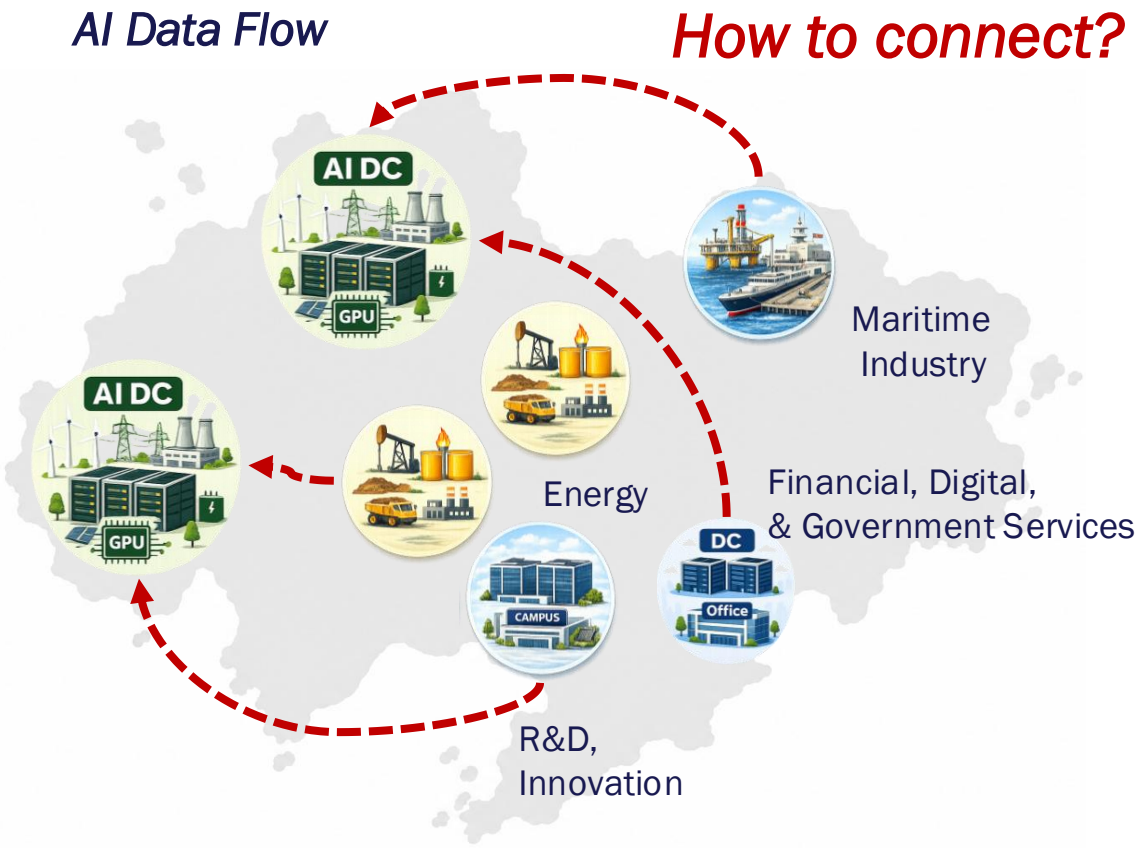
What is AICC

- Distributed cloud data centers and **regional sites** interconnected into a **unified accelerated** computing space
- Add AI computing capabilities to regional industries and societies



Connectivity Issue

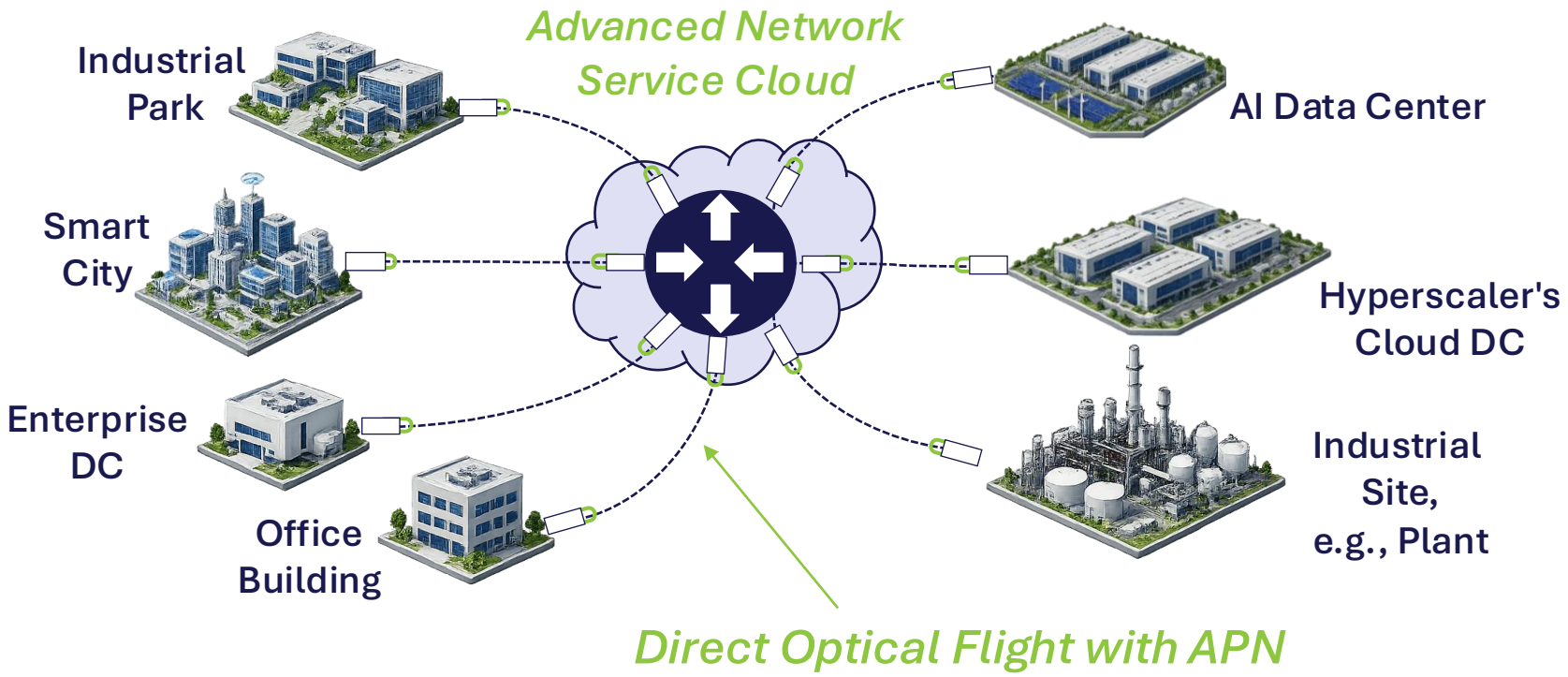
How can we make such hyperscale AI data centers efficiently accessible from industrial users?



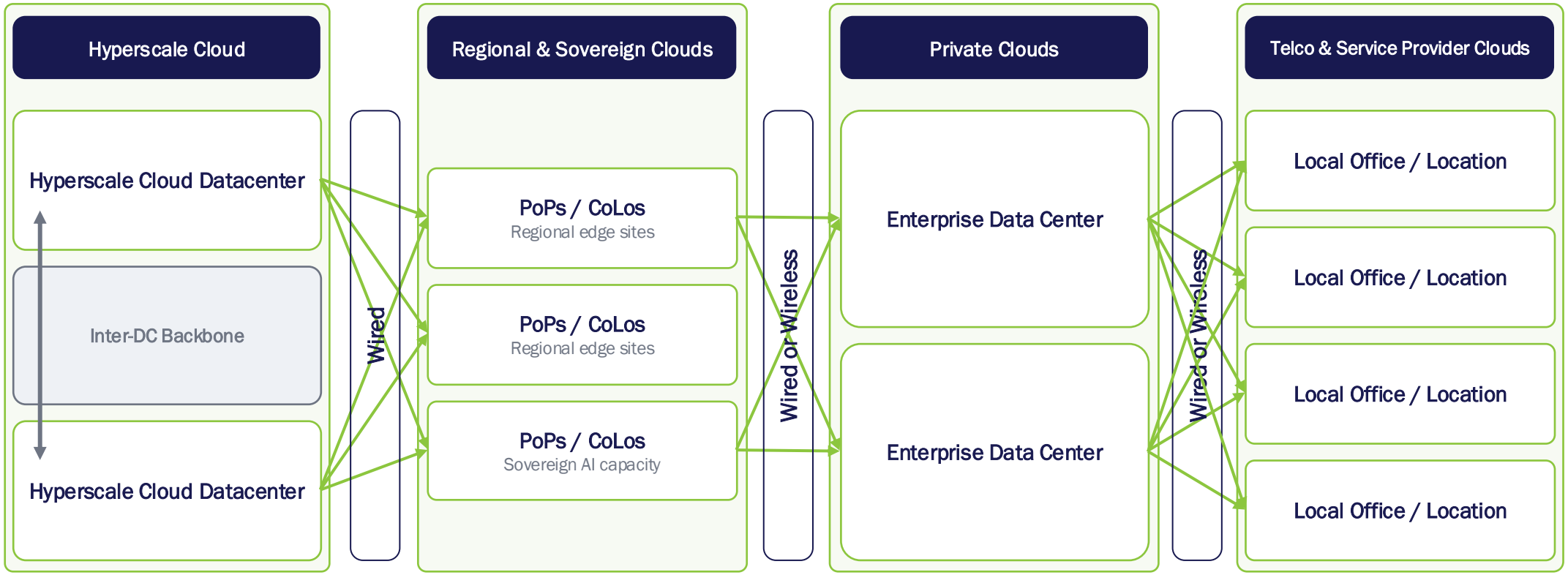
Scale is not everything. Connectivity/Reachability is equally important!

Hub-and-Spoke Networking with Open APN

APN efficiently connects **geographically distributed** sites to a hub through **direct optical flights**, forming a **Hub-and-Optical-Spoke** topology.



AI Computing Continuum Scale Out Span



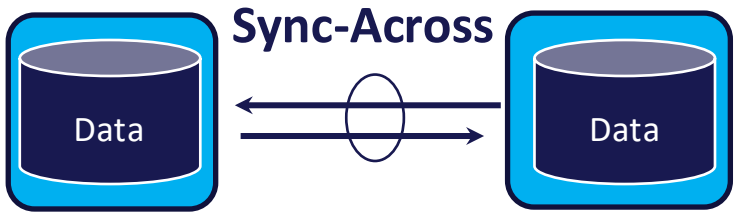
Breaking the Data Center Walls: Unifying AI Fabrics for Agentic AI at the Edge

Tuesday, October 13, 2026 | 1:40 PM PDT | SJCC, Concourse Level, 211 | Prakash Jain (Celestica)

AICC Use Cases

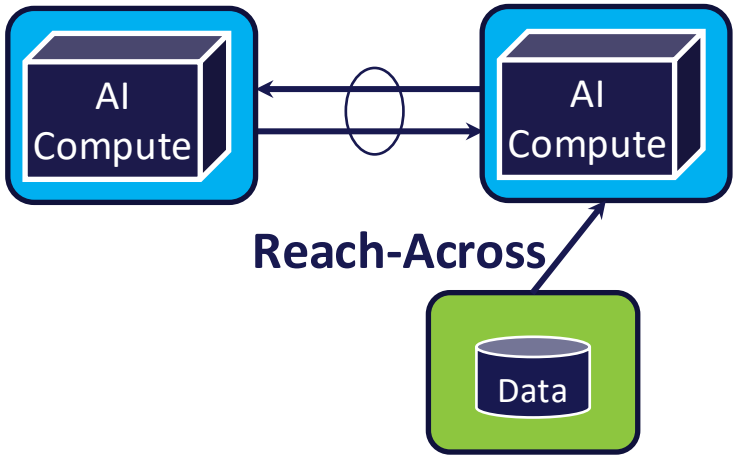
Service Industries

e.g., Financial DCs



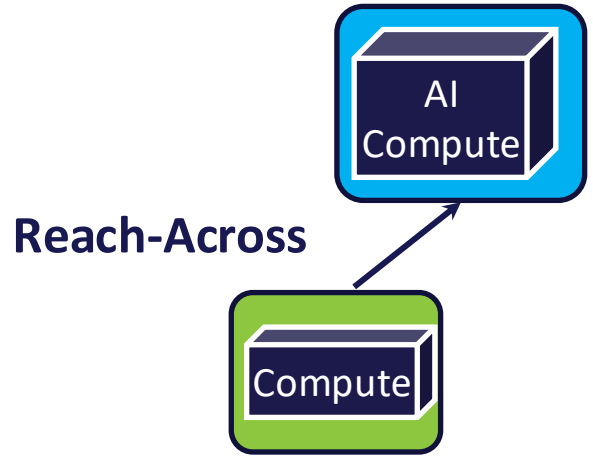
Sovereign/Industrial AI

Scale-Across (at Geo-Scale)



Physical AI

e,g, Logistics Warehouses



AI Computing Continuum Use Cases and Collaboration Items | Tuesday, October 13, 2026 | 1:45 PM PDT | SJCC, Concourse Level, 211

Naoto Umemori (NTT DATA) | Hidetsugu Sugiyama (Red Hat) | Xiaojing Zhang (NTT DOCOMO Business) | Junichi Sugiyama (1Finity) | Koji Watanabe (NTT)

OCP and IOWN Global Forum Complete The Stack

IT Hardware Suppliers

DC Facilities

Computer

Storage

Network

NIC

Adaptor

OPEN Compute Project®

Open Hardware Product Specifications

Open and Modular Designs

IOWN GLOBAL FORUM™

Full-Stack, End-to-End Reference Designs for Solution Systems

Vertical Use Cases (Problems, Solutions, and ROI Estimates)

Forum's Innovation Thinking, and Forum's Technology Architecture

Adopters

Infrastructure Service Providers

Financial Service Institutes

Construction Property Developers

Manufacturers Plants

Logistics Warehouses

Broadcasters

Specification Targets For Next Year

OPEN Compute Project®

Open Hardware Product Specifications

Open and Modular Designs

Hyperscale Adjacent	OpenAPN Networking	AI-Native Edge & Software	AI-Native Wireless
Hyperscale Adjacent Rack Specification(s)	OpenAPN NIC based on OCP NIC specification	AI Native Edge-MHS specification revisions	AI Native Wireless AuRu specification
Hyperscale Adjacent Server Specification(s)	OpenAPN SmartNIC based on OCP Open Adapter Card or OCP NIC specification	OCP Software Initiative	AI Native Wireless Bridge Board specification leveraging OCP Open Adapter Card
			AI Native Wireless Experience Center at UNH

Anticipated OCP contributions over the next 12 months

Bridging the AI Continuum: Trust, Orchestration, and Open Standards

Tuesday, October 13, 2026 | 4:35 PM PDT | SJCC, Concourse Level, 211

Steven Yates (Federant) | Raúl Álvarez (OCP) | Vincent Nguyen (HPE) | Dirk Blevins (Intel) | Çağatay Yilmaz (RISE)

Priority for the Next Year and Call to Action

Priority for the Next Year

- Develop a few compelling combinations of IOWN Global Forum use cases and OCP product specifications that drive the early deployment of AICC.

Call to Action: Let's start with case studies

- Review AICC use cases being developed at IOWN Global Forum
  - Several use cases are presented in today’s session.
- Identify OCP product specifications, e.g., AI Native Edge-MHS, to support AICC use cases

Thank You!

